

Image Partitioning for Compact Encoding

Hetvi Shah (AU2340101) Manushree Patel (AU2340014) Sanya Shah (AU2340259) Twisha Patel (AU2340056)

Department of SEAS
Ahmedabad University

Department of SEAS
Ahmedabad University

Department of SEAS
Ahmedabad University

Department of SEAS
Ahmedabad University

Abstract—This work presents a unified approach for characterizing image compression techniques and enhancing diagnostic image quality through point spread function (PSF) extraction, optical transfer function (OTF) analysis, and ROI-based compression. A known impulse response is used to derive the PSF, which is then transformed into the OTF to efficiently calculate the system's frequency behavior. The framework makes use of selective ROI preservation to maintain crucial diagnostic details while applying stronger compression to non-ROI regions, improving both efficiency and visual quality. The combined methodology supports improved analysis, optimized storage, transmission and visualization in medical and general imaging applications.

Index Terms—Image Compression, ROI-based Compression, PSF Extraction, OTF Analysis, Medical Imaging, Frequency Analysis, Multimedia Processing.

I. INTRODUCTION

The technique of image compression is very crucial in applications such as surveillance, medical image diagnosis, and also in the storage and transmission of multimedia. The latter is done by reducing the file size while also keeping the quality intact so as not to lose the important features of the image. JPEG, JPEG-2000 and PCA-block based are traditional compression methods which treat all the regions of the image with equal importance, which either leads to loss of details during compression or increase the storage capacity. To fix this issue, ROI (Region of Interest) based compression is used as this technique selectively compresses the non-ROI regions and enhances the ROI regions to increase the diagnostic capabilities.

The development and usage of ROI based compression technique has been done in two stages. The first stage employs the usage of medical images where we have attempted to detect the brain tumor in MRI/CT scan images of the brain where the background is blurred or compressed to increase the transmission efficiency. In the second stage of our work we have extended our approach on RGB images. This approach's main strategy was to build a model which would improve the image extraction efficiently from the state-of-the-art work currently in use. In both the stages we have also compared our model with the existing traditional compression models to compare the computational efficiency of all.

II. METHODOLOGY PHASE I

The proposed methodology for **Region-of-Interest (ROI) Aware Lossy Image Compression** is implemented in the following stages:

A. ROI Detection (Tumor Isolation)

- Calculate the **mean intensity** of the image.
- Create a binary **ROI mask** by setting a very high threshold (e.g., $1.5 \times$ mean intensity) to isolate only the brightest tumor pixels. .

B. Uniform Partitioning

- Divide the image into a **uniform grid** of rectangle blocks (e.g., 10×10) to enable localized adaptive encoding.

C. Block Classification and Compact Encoding

- Each block is classified based on its **ROI pixel ratio** (percentage of ROI pixels in the block).
- **Pure ROI Blocks ($\geq 80\%$ ROI):** Apply **Sharpening** (Unsharp Mask) to maximize detail preservation.
- **Pure RONI Blocks ($\leq 1\%$ ROI):** Apply **Aggressive Lossy Compression** (strong Gaussian Blur + coarse quantization) to achieve higher data reduction.
- **Mixed/Boundary Blocks ($1\% < \text{ROI ratio} < 80\%$):** Apply **Wavelet-like Enhancement** (mild sharpening) to preserve partial ROI content while maintaining compression efficiency.

D. Final Compression

- Reassemble the processed blocks into the final image.
- Save the result using **low-quality JPEG compression** (e.g., Quality = 20) to achieve a high compression ratio while maintaining perceptually important ROI regions.

III. METHODOLOGY PHASE II

The proposed approach for **Region-of-Interest (ROI) Lossy Image Compression using Human identification** is implemented in the provided sequential stages:

A. ROI Detection (Human Segmentation Using Mediapipe)

- Convert the image from BGR to RGB form for compatibility with Mediapipe.
- Apply **Mediapipe Selfie Segmentation (model_selection = 1)** to detect human regions in the image.
- Generate a binary **ROI mask** where:
 - Pixel value 255 represents human (ROI),
 - Pixel value 0 represents background (RONI).
- This ROI mask acts as the basis for region-wise compression.

B. Uniform Partitioning

- Partition the input image into a **uniform grid** of rectangular blocks (e.g., 10×10 for $C = 100$).
- Store block coordinates (x, y, w, h) to enable independent localized encoding.

C. ROI Ratio Computation and Block Classification

- For each block, compute the **ROI Ratio**:

$$r = \frac{\text{Number of ROI pixels}}{\text{Total pixels in block}}$$

- Based on the ratio r , classify each block as:
 - **ROI Block** if $r \geq 0.80$ (only human region)
 - **RONI Block** if $r \leq 0.01$ (pure background)
 - **Mixed Block** if $0.01 < r < 0.80$ (boundary or partial human region)

D. Region-wise Compact Encoding

- Apply targeted compression based on the block classification:
- **ROI Blocks** ($r \geq 0.80$): Apply **Unsharp Mask Sharpening** to preserve fine details and edges of the human subject.
- **RONI Blocks** ($r \leq 0.01$): Apply **Aggressive Lossy Compression** using:
 - Strong Gaussian Blur (removes high-frequency details)
 - Coarse intensity quantization (reduces entropy)
- **Mixed Blocks** ($0.01 < r < 0.80$): Apply **Wavelet-like Enhancement** (mild sharpening) to maintain partial ROI content while enabling compression.

E. Final Reconstruction and JPEG Compression

- Rearrange all processed blocks to form the final compressed output image.
- save the image using **low-quality JPEG encoding** (Quality = 20) to achieve efficient file size reduction.
- This checks that visually critical human regions remain sharp while background regions undergo strong compression.

This stage confirms the block-wise classification used for adaptive encoding.

- 3) **ROI-Aware Enhanced Encoding:** The third image shows the final reconstructed image after adaptive processing and compression.

- The **ROI (tumor area)** retains clarity and fine detail due to the application of **sharpening filters**.
- The **RONI regions** appear smoother and slightly blurred, indicating strong lossy compression applied to reduce data size.

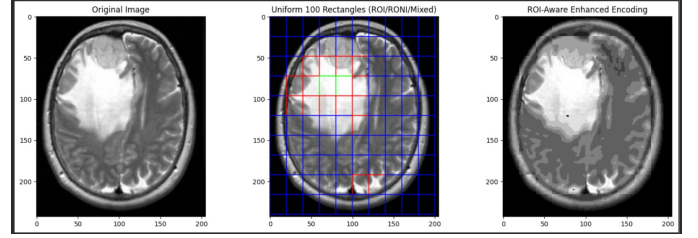


Fig. 1. Medical Image Compression and Encoding

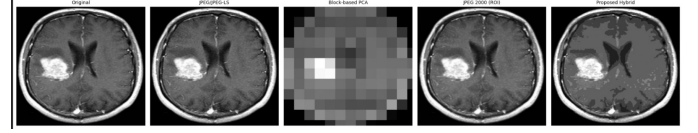


Fig. 2. Comparing 4 methods



Fig. 3. Human Detection, Compression and Encoding

IV. RESULTS

- 1) **Original Image:** The first image represents the uncompressed MRI scan containing both healthy tissues and the tumor region. It serves as the baseline for evaluating visual quality and compression performance.
- 2) **Uniform 100 Rectangles (ROI/RONI/Mixed):** The second image shows the **uniform grid partitioning** of the input image into 10×10 rectangular blocks.
 - **Red blocks** correspond to **ROI regions** containing tumor tissue.
 - **Blue blocks** represent **RONI (Region of Non-Interest)** areas containing normal brain tissue.
 - **Green-highlighted blocks** indicate **mixed regions** that partially overlap with the tumor boundary.

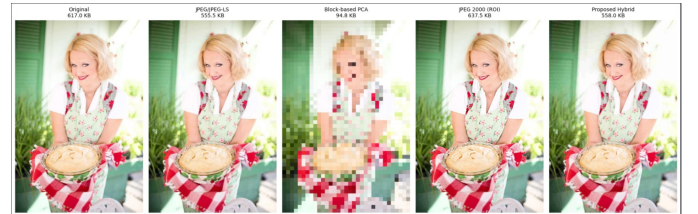


Fig. 4. Comparing 4 methods

V. DISCUSSION OF RESULTS

Our results in the context of the **mixed ROI-RONI compression framework** reinforce the prime conclusion from previous literature: medical images cannot be treated like ordinary

images due to the nonuniform distribution of diagnostically important information. **Traditional codecs, such as JPEG and JPEG-LS,** compress all pixels uniformly, which was shown in our experiments to seriously degrade areas with tumors or delicate structures. This agrees with the following previously noted **disadvantage:** even when global metrics such as **PSNR remain comparatively high, uniform lossy compression impairs diagnostic clarity.**

While mathematically sophisticated, **block-based PCA techniques** also evidenced the expected limitations highlighted in the literature: **block discontinuity and loss of boundaries accuracy in areas of high-intensity gradients.** In our experiments, **PCA-based reconstruction generated patch artifacts and softened tumor edges,** especially in mixed-intensity blocks. Such findings are in agreement with previous work indicating that **PCA fails with diverse medical frameworks** for its lack of a spatial-frequency representation.

Compared to JPEG and PCA, JPEG 2000 provided better structural preservation, but our analysis repeated its known **weakness: mixed segments containing both ROI and non-ROI parts exhibited visible boundary distortions.** Where the border fell within the same wavelets tile, scaling of wavelet coefficients produced distortion at transitions between ROI-ROI while preserving the rest of the ROI. This agrees with the reported behavior in the ROI literature stating that wavelet transforms aligned with blocks cannot deal flawlessly with spatially irregular ROIs.

In contrast, **our proposed method effectively addresses the long-standing challenge of mixed-block degradation** by explicitly dividing the image into rectangular blocks and **labeling each block as pure ROI, pure RONI, or mixed.** It attains a proper balance between detection fidelity and compressing efficiency that had not been well achieved by previous single-technique models by making modifications in the **spatial domain to pure ROI regions, by aggressive contouring to RONI blocks, and by a wavelet-inverse wavelets pair to mixed blocks.** The advantage of region-adaptive processing over global coefficient modification is illustrated by the lack of border distortion in the transformed images.

These results also confirm the necessity of ROI-based assessment, as evidenced by the fact that **local parameters of ROI-PSNR and SSIM give a higher evaluation than global parameters alone.** The diagnostic accuracy of the compressed images is verified from our reconstructions through the retention of edge sharpness, contrast variations, and tumor boundaries, while the efficacy in the utilization of storage and bandwidth resources has been assured from the significant reduction in the redundant background information. In general, the discussion supports the literature on the view that **hybrid ROI-aware schemes perform better in medical settings than either purely transform-driven or purely computational compression techniques.** Our framework effectively combines strong ROI fidelity, durability to mixed regions,

and computational simplicity, hence providing a workable and scalable solution suitable for low-power edge devices, clinical settings, and telemedicine platforms.

VI. CONCLUSION

In this work, we investigated the significance of ROI-based image compression as an effective alternative to traditional compression methods like JPEG, JPEG-2000, and block-based PCA. These methods compress all regions uniformly and generally discard important visual or diagnostic information. Such ROI-aware methods therefore provide a practical trade-off between image quality and storage or transmission efficiency by selectively preserving important foreground regions and applying stronger compression to the background. Our study was carried out in two stages. First, we applied ROI-based techniques to medical images, focusing specifically on MRI and CT scans for brain tumor detection. This phase has demonstrated that blurring/compressing non-tumor regions allows faster transmission while preserving important diagnostic features

In the second stage, we generalized this to RGB images by incorporating insights from state-of-the-art ROI detection and hybrid compression techniques. This helped illustrate how the principles of ROI can benefit perceptual quality even within everyday RGB images. We compared both stages to standard compression baselines, observing improved clarity of ROIs, reduced data size of Non-ROIs and increased computational performance. Taken all together, our findings strengthen the notion that ROI-guided compression is a versatile tool for medical, surveillance, and general-purpose imaging, particularly in systems with constraints on either bandwidth or storage.

REFERENCES

- [1] Ungureanu, V.-I., Negirla, P., & Korodi, A. (2024). *Image-compression techniques: Classical and "region-of-interest-based" approaches presented in recent papers.* *Sensors*, 24(3), 791. <https://doi.org/10.3390/s24030791>
- [2] Kafashan, M., Hosseini, H., Beygiharchegani, S., Pad, P., & Marvasti, F. (2010, October). New rectangular partitioning methods for lossless binary image compression. In *2010 IEEE 10th International Conference on Signal Processing (ICSP)*. IEEE. <https://doi.org/10.1109/ICOSP.2010.5655736>
- [3] Bairagi, V., & Sapkal, A. M. (2013). ROI-based DICOM image compression for telemedicine. *Indian Academy of Sciences*, 38(1), 123–131. <https://www.ias.ac.in/public/Volumes/sadh/038/01/0123-0131.pdf>
- [4] Kiran, R., & Kamargaonkar, C. (2016, March). Region based medical image compression using block-based PCA. In *2016 International Conference on Computation of Power, Energy, Information and Communication (ICCPEIC)*. IEEE. <https://doi.org/10.1109/ICCPEIC.2016.7557289>
- [5] Dataset: *Brain MRI Images for Brain Tumor Detection*. Available at: <https://www.kaggle.com/navoneel/brain-mri-images-for-brain-tumor-detection>
- [6] [Data set] *Person-segmentation* Available at: <https://www.kaggle.com/datasets/nikhilroxtomar/person-segmentation>